

Shih-Yi (Matteo), Soo

Homepage: matteosoo.github.io ◇GitHub: github.com/matteosoo
Phone: (+886) 981-359-060 ◇Email: matteosoo11@gmail.com

SUMMARY

Senior Machine Learning Engineer with 5+ years of experience in NLP and GenAI systems. Specialized in production-grade RAG and agentic applications on GCP. Led Taiwan’s first financial conversational AI from PoC to launch (July 2025); currently serving as project tech lead and TPM for an internal agentic chatbot for wealth management.

WORK EXPERIENCE

- E.Sun Financial Holding Co., Ltd**
Senior Machine Learning Engineer

Taipei, Taiwan
Mar 2023 - Present
- **Agentic Wealth Management Chatbot:** Served as project tech lead and TPM (timeline, requirement alignment); shipped an internal agentic chatbot for wealth management RMs in 3 months (Q2 2026) — 2,000+ page enterprise KM with adaptive retrieval and query reformulation (hit rate 90% → 92%).
 - ↔ **Architecture:** Architected on Google ADK with modular workflow skills; LLM orchestrates multi-step reasoning while deterministic guardrails (retrieval-count cap via `tool_context` state) enforce stable execution and prevent runaway loops.
 - **RAG Chatbot & Cloud Infrastructure:** Led Taiwan’s first conversational GenAI service in finance (PoC → launch July 2025); built a dual-track system (rule-based for compliance decisions, RAG for market Q&A) with hybrid retrieval, VLM-based ETL, and automated SLA alerting; deployed on serverless computing platform - Cloud Run. Demoed at **NVIDIA GTC 2025 & FinTech Taipei 2024**; featured in 50+ media outlets.
 - **AI Community:** Organized Taiwan’s first RAG Hackathon (487 teams); obtained AI new-type and invention patents for GPT automated evaluation methodology.

- E.Sun Financial Holding Co., Ltd**
Machine Learning Engineer

Taipei, Taiwan
Jul 2021 - Mar 2023
- **Speech (TTS):** Built in-house TTS service (RTF 0.3) using Tacotron2/VITS with 20-hour custom dataset; fine-tuned IPA-based models to support 4-language interleaving for telephony IVR systems.
 - **BERT Chatbot & On-premise Infrastructure:** Managed 12+ BERT-based chatbots on Kubernetes; deployed gateway APIs handling 1,000+ monthly calls.

- Walsin Lihwa Corporation**
Software Engineer - Internship

Taipei, Taiwan
Apr 2018 - Jul 2018
- Applied PageRank and bi-LSTM for industrial text summarization; automated HR web scraping for talent acquisition.

TECHNICAL SKILLS

Generative AI	LLM, RAG, Agentic Orchestration, Google ADK, PyTorch
NLP & Speech	BERT, Text-to-Speech (TTS), Text Mining, IPA Encoding
Cloud & Infra	GCP (Cloud Run, AlloyDB, BigQuery, Vertex AI), Kubernetes, Docker, Airflow
Dev Tools	Python, FastAPI, Git, SQL, Linux

EDUCATION

National Tsing Hua University	2018 - 2021
M.S. in Computer Science	Hsinchu, Taiwan
Soochow University	2014 - 2018
B.S. in Computer Science and Information Management	Taipei, Taiwan